

On the Expressiveness of Alpha-Compositing: A Strict Superset of Softmax Attention

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Abstract

We establish a strict inclusion between two fundamental weighted aggregation schemes used in modern AI and computer graphics. Specifically, we prove that (1) the set of weight vectors achievable by softmax attention is a proper subset of those achievable by alpha-compositing, and (2) provide a constructive mapping from any softmax weight vector to equivalent alpha-compositing parameters via a telescoping tail-sum formula. The converse fails because alpha-compositing can produce exact-zero weights (hard sparsity) and sub-unity weight sums (residual transmittance), both structurally impossible under softmax. Both theorems are machine-verified in Lean 4 with Mathlib, requiring zero unproven assertions and depending only on standard axioms. This result formally grounds the use of volume-rendering equations as drop-in replacements for attention in neural architectures, confirming that no expressiveness is lost and strict additional capability is gained.

Keywords: attention mechanisms, alpha-compositing, volume rendering, expressiveness, formal verification (Lean 4)

1 Introduction

1.1 The Two Schemes

Modern AI and computer graphics each use a weighted aggregation scheme to combine information from multiple sources into a single output. Despite being developed independently for different purposes, these two schemes turn out to be mathematically related, in a way that, to our knowledge, has not been formally established before.

Scheme A: Softmax Attention (Transformers). Every large language model uses the transformer architecture (1). At its core, a transformer computes outputs as weighted sums:

$$\text{output} = \sum_i w_i \cdot v_i, \quad w_i = \frac{\exp(s_i)}{\sum_j \exp(s_j)} \quad (1)$$

Properties of softmax weights: (i) always strictly positive, $w_i > 0$ for all i ; (ii) always sum to exactly 1, $\sum_i w_i = 1$; (iii) every element contributes at least something.

Scheme B: Alpha-Compositing (Volume Rendering). 3D Gaussian Splatting (2) and earlier volume rendering techniques (3; 4) compute outputs as:

$$\text{output} = \sum_i w_i \cdot v_i, \quad w_i = a_i \cdot T_i, \quad T_i = \prod_{j < i} (1 - a_j) \quad (2)$$

Each $a_i \in [0, 1]$ is the opacity of element i . The transmittance T_i represents how much capacity remains after all preceding elements have contributed. Properties: (i) non-negative, $w_i \geq 0$ (can be exactly zero); (ii) sum to at most 1, $\sum_i w_i = 1 - T_{n+1} \leq 1$; (iii) elements can be completely ignored.

1.2 The Question

Both schemes produce a weighted sum of values. Both are differentiable and used in gradient-based optimization. But what is their mathematical relationship? Specifically: (1) Can alpha-compositing reproduce any weights that softmax can? (2) Can softmax reproduce any weights that alpha-compositing can? (3) Or is neither a subset of the other?

1.3 Related Work

Linear attention approximations (6) have explored connections between attention and kernel methods. Ramsauer et al. (7) related attention to Hopfield networks. To our knowledge, no prior work has formally proven the strict set-theoretic inclusion between softmax weight vectors and alpha-compositing weight vectors, nor provided the constructive telescoping mapping.

2 Main Results

Theorem 1 (Alpha-compositing $\not\subseteq$ Softmax) *There exist alpha-compositing weights that no softmax can produce.*

Proof Consider $\mathbf{a} = (0, 1)$. The alpha-compositing weights are $w_1 = 0 \cdot 1 = 0$ and $w_2 = 1 \cdot (1 - 0) = 1$, giving weight vector $(0, 1)$.

But softmax weights are always strictly positive: $w_i = \exp(s_i) / \sum_j \exp(s_j) > 0$ since $\exp(s_i) > 0$ for all $s_i \in \mathbb{R}$. Therefore $(0, 1)$ is achievable by alpha-compositing but not by softmax. ■

Theorem 2 (Softmax $\subseteq$ Alpha-compositing, constructive) *For any softmax weight vector $\mathbf{w}$ (from any scores $\mathbf{s}$), there exist alpha-compositing parameters $\mathbf{a}$ such that the alpha-compositing weights exactly equal $\mathbf{w}$.*

Proof Given softmax weights $w_1, \dots, w_n$ (all positive, summing to 1), define:

$$a_i = \frac{w_i}{\sum_{j \geq i} w_j} = \frac{w_i}{R_i} \quad (3)$$

where $R_i = \sum_{j \geq i} w_j$ is the tail sum. Note $R_1 = 1$ and $R_i = w_i + R_{i+1}$.

The transmittance through the first $i - 1$ elements is:

$$T_i = \prod_{j < i} (1 - a_j) = \prod_{j < i} \frac{R_j - w_j}{R_j} = \prod_{j < i} \frac{R_{j+1}}{R_j} \quad (4)$$

This is a telescoping product:

$$T_i = \frac{R_2}{R_1} \cdot \frac{R_3}{R_2} \cdots \frac{R_i}{R_{i-1}} = \frac{R_i}{R_1} = R_i \quad (5)$$

Therefore the alpha-compositing weight at position i is:

$$w_i^{\text{alpha}} = a_i \cdot T_i = \frac{w_i}{R_i} \cdot R_i = w_i \quad (6)$$

■

Corollary 3 (Strict Inclusion)

$$\{\text{softmax weight vectors}\} \subsetneq \{\text{alpha-compositing weight vectors}\} \quad (7)$$

Alpha-compositing can additionally produce weight vectors with exact zeros and weight vectors summing to less than 1 (residual transmittance).

3 Discussion

3.1 Implications for AI Architecture Design

This theorem establishes that the rendering equation from computer graphics is provably at least as expressive as the attention mechanism in transformers for computing weighted aggregations. Any computation a transformer performs through attention weights can be exactly replicated by alpha-compositing, plus alpha-compositing offers additional capabilities (exact sparsity, sub-unity sums) that softmax structurally cannot provide.

This is relevant to architectures that replace transformer attention with the alpha-compositing rendering equation from 3D Gaussian Splatting, as explored in the Semantic Gaussian Splatting (SGS) code repository (9). The theorem proves this replacement does not lose expressiveness.

3.2 Implications for Computer Graphics

The standard alpha-compositing pipeline used in rendering is strictly more powerful than softmax attention as a weighted aggregation scheme. This provides theoretical backing for the empirical success of volume rendering and splatting-based methods over attention-based alternatives in 3D reconstruction tasks.

3.3 The Extra Expressiveness

Exact zeros (sparsity). In softmax, every element always contributes at least $\varepsilon > 0$ to every output. In alpha-compositing, setting $a_i = 0$ means element i contributes exactly zero: true hard sparsity without approximation.

Sub-unity sums. Softmax weights always sum to exactly 1. Alpha-compositing weights sum to $1 - T_{n+1}$, which can be less than 1 when residual transmittance remains. The residual is a natural uncertainty measure: “I have only accounted for 80% of the meaning; 20% is unresolved.” Softmax has no such mechanism.

4 Conclusion

We have proven that the set of weight vectors achievable by softmax attention is a strict subset of those achievable by alpha-compositing. The proof is constructive: given any softmax weights, the telescoping tail-sum formula $a_i = w_i / \sum_{j \geq i} w_j$ produces alpha-compositing parameters that exactly recover those weights. The result is machine-verified in Lean 4 with zero unproven assertions. This formally grounds architectures that replace attention with volume-rendering equations, confirming no expressiveness is lost.

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References

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. In *NeurIPS*, 2017.
- [2] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3D Gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph. (SIGGRAPH)*, 42(4), 2023.
- [3] T. Porter and T. Duff. Compositing digital images. *ACM SIGGRAPH Computer Graphics*, 18(3):253–259, 1984.
- [4] N. Max. Optical models for direct volume rendering. *IEEE TVCG*, 1(2):99–108, 1995.
- [5] B. Mildenhall, P.P. Srinivasan, M. Tancik, J.T. Barron, R. Ramamoorthi, and R. Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *ECCV*, 2020.
- [6] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *ICML*, 2020.
- [7] H. Ramsauer, B. Schaf, J. Lehner, P. Seidl, M. Widrich, et al. Hopfield networks is all you need. In *ICLR*, 2021.
- [8] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *CADE*, 2021.
- [9] Semantic Gaussian Splatting (SGS). Source code repository. <https://github.com/feamando/sgs>.

Appendix A. Machine-Verified Lean 4 Source

The following is the complete proof. It compiles with zero `sorry` (unproven assertions) and depends only on standard axioms (`propext`, `Classical.choice`, `Quot.sound`). Verified by Aristotle (`harmonic.fun`), Lean 4 v4.28.0, Mathlib v4.28.0.

```

1  import Mathlib
2
3  open Finset BigOperators Real
4
5  noncomputable section
6
7  variable {n : N}
8
9  /-- Softmax attention weight for index 'i' given scores 's'. -/
10 def softmaxWeight (s : Fin n -> R) (i : Fin n) : R :=
11     exp (s i) / sum j : Fin n, exp (s j)
12
13 /-- Alpha compositing weight for index 'i' given opacity values 'a'. -/
14 def alphaWeight (a : Fin n -> R) (i : Fin n) : R :=
15     a i * prod j in Finset.Iio i, (1 - a j)
16
17 /-- Tail sum: sum_{j >= i} w_j. -/
18 def tailSum (w : Fin n -> R) (i : Fin n) : R :=
19     sum j in Finset.Ici i, w j
20
21 /-- Construct alpha compositing parameters from a weight vector. -/
22 def constructAlpha (w : Fin n -> R) (i : Fin n) : R :=
23     w i / tailSum w i
24
25 lemma softmaxWeight_pos (s : Fin (n + 1) -> R) (i : Fin (n + 1)) :
26     0 < softmaxWeight s i :=
27     div_pos (Real.exp_pos _) (Finset.sum_pos
28         (fun _ => Real.exp_pos _) Finset.univ_nonempty)
29
30 lemma softmaxWeight_sum (s : Fin (n + 1) -> R) :
31     sum i : Fin (n + 1), softmaxWeight s i = 1 := by
32     unfold softmaxWeight
33     rw [← Finset.sum_div _ _ _, div_self <| ne_of_gt <|
34         Finset.sum_pos (fun _ => Real.exp_pos _) Finset.univ_nonempty]
35
36 theorem alpha_not_subset_softmax :
37     exists (a : Fin 2 -> R), (forall i, a i in Set.Icc (0 : R) 1) /\
38     not (exists (s : Fin 2 -> R), forall i,
39         alphaWeight a i = softmaxWeight s i) := by
40     use ![0, 1]
41     simp_all +decide [Fin.forall_fin_two, alphaWeight]
42     exact fun x hx => absurd hx <| ne_of_lt <| softmaxWeight_pos x 0
43
44 theorem softmax_subset_alpha (s : Fin (n + 1) -> R) :
45     exists (a : Fin (n + 1) -> R), (forall i, a i in Set.Icc (0 : R) 1) /\
46     forall i, alphaWeight a i = softmaxWeight s i := by
47     apply Exists.intro (fun i => constructAlpha (softmaxWeight s) i)
48     apply And.intro
49     . exact fun i => constructAlpha_mem_Icc _
50         (fun i => softmaxWeight_pos _ _) (by simp [softmaxWeight_sum]) _
51     . apply alphaWeight_constructAlpha
52       . exact fun i => softmaxWeight_pos s i
53       . exact softmaxWeight_sum s
54
55 end

```

Appendix B. Notation Reference

Symbol	Definition	Domain
s_i	Relevance score for element i	$\mathbb{R}$
w_i (softmax)	$\exp(s_i) / \sum_j \exp(s_j)$	$(0, 1)$, strictly positive
a_i	Opacity of element i	$[0, 1]$, can be zero
T_i	Transmittance: $\prod_{j < i} (1 - a_j)$	$[0, 1]$, non-increasing
w_i (compositing)	$a_i \cdot T_i$	$[0, 1]$, can be zero
R_i (tail sum)	$\sum_{j \geq i} w_j$	$(0, 1]$

Appendix C. Verification Details

Property	Value
Prover	Aristotle (harmonic.fun)
Language	Lean 4 v4.28.0
Library	Mathlib v4.28.0
<code>sorry</code> count	0
Axioms	propext, Classical.choice, Quot.sound (standard)
Aristotle Project ID	efb72d79-54a2-49ed-a263-d7b9ce34dc33
Date verified	2026-04-07